Jenkins CI/CD Pipeline Tomcat-Apache : Deployment & Testing on Linux Server

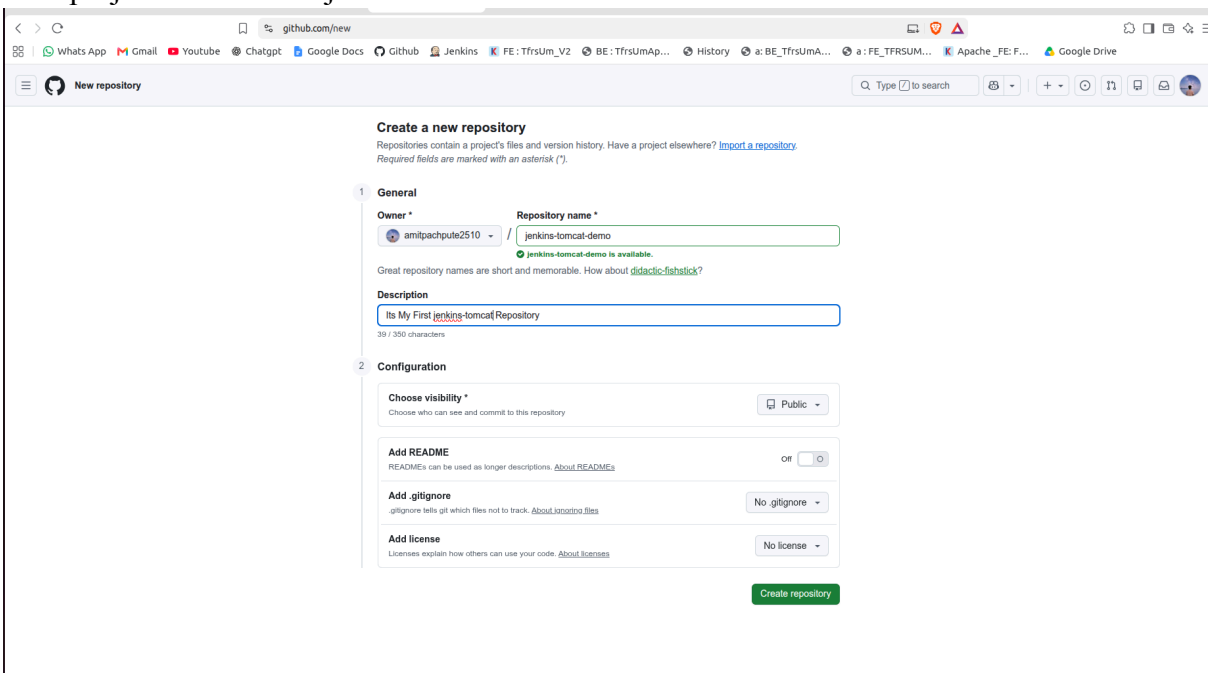
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Step 1: Create a GitHub repository

Go to Your Github Accont and Login
Then Create a GitHub repository

- Repository name: jenkins-tomcat-demo
- Visibility: Public (simplest for learning)
- Add a README: Optional
- Click Create repository.

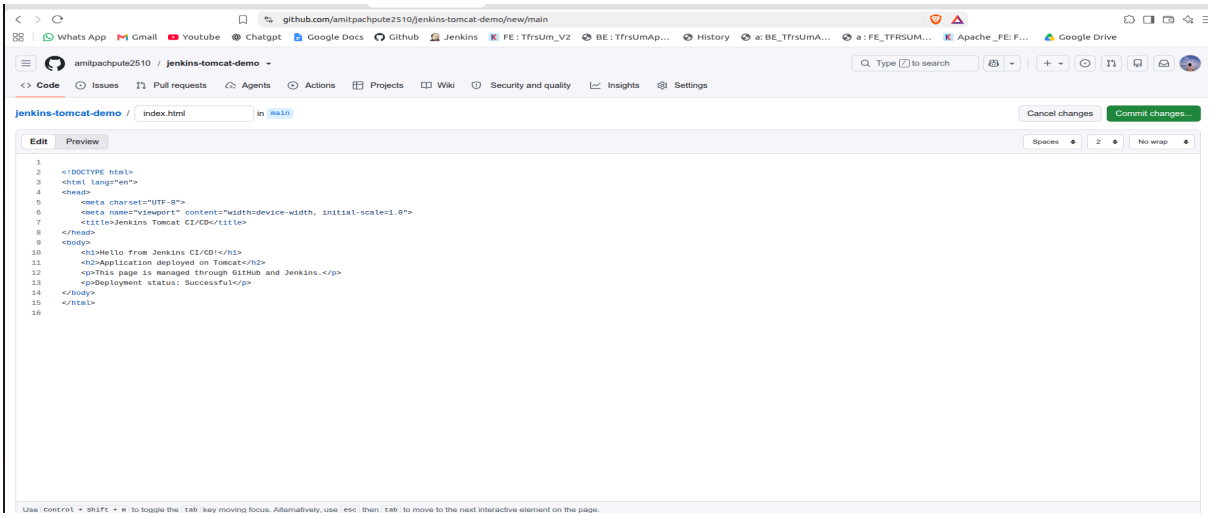
Your project will contain just one website file:



Create index.html on GitHub using Add file → Create new file and paste

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Jenkins Tomcat CI/CD</title>
</head>
<body>
```

```
    <h1>Hello from Jenkins CI/CD!</h1>
    <h2>Application deployed on Tomcat</h2>
    <p>This page is managed through GitHub and Jenkins.</p>
    <p>Deployment status: Successful</p>
</body>
</html>
```



Go back to GitHub and create a second file named exactly `Jenkinsfile` — no `.txt` or `.groovy` extension.

Paste this pipeline:

```
pipeline {
    agent any

    options {
        disableConcurrentBuilds()
        timestamps()
    }

    triggers {
        // Check GitHub for new commits approximately every 2 minutes
        pollSCM('H/2 * * * *')
    }

    stages {
        stage('Checkout from GitHub') {
            steps {
                checkout scm
            }
        }

        stage('Validate HTML') {
            steps {
                sh '''
```

```
                set -eu

                test -s index.html
                grep -qi '<html' index.html

                echo "HTML validation passed."
            '''
        }
    }

    stage('Build Backup') {
        steps {
            sh '''
                set -eu

                sudo /usr/local/sbin/deploy-jenkins-demo "$WORKSPACE"
            '''
        }
    }

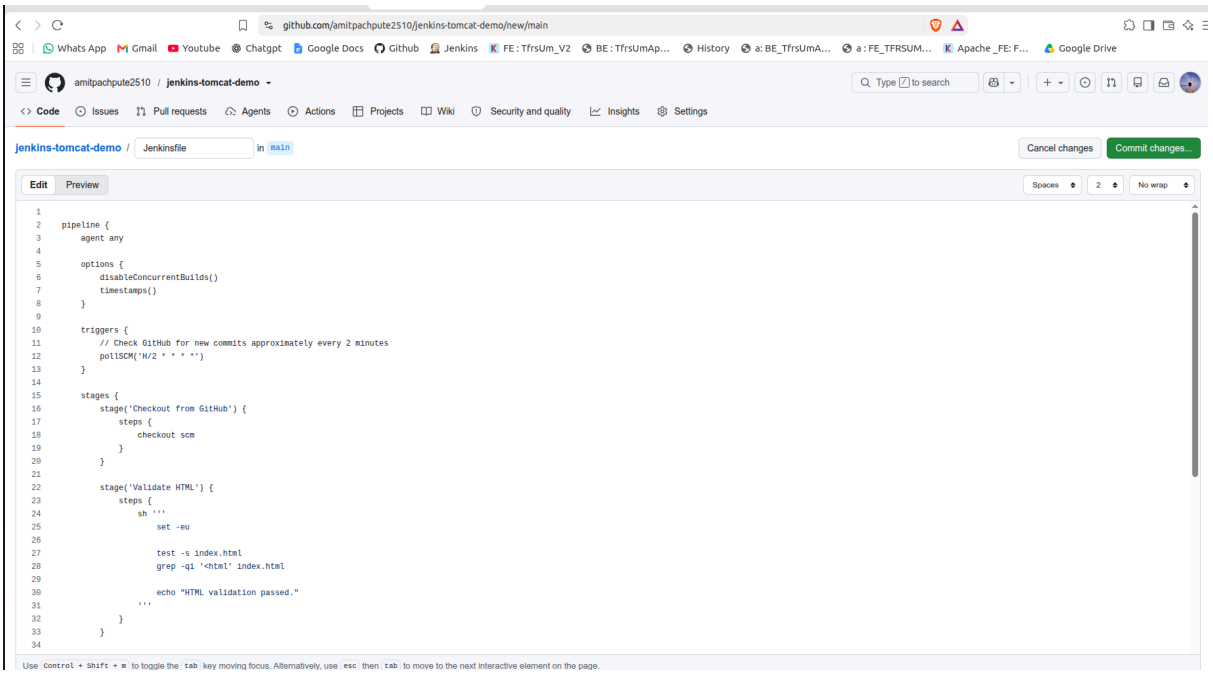
    stage('Verify Deployment') {
        steps {
            sh '''
                set -eu

                test -s /opt/apache-tomcat-10.1.56/webapps/jenkins-demo/index.html

                echo "Tomcat deployment file exists."
            '''
        }
    }
}

post {
    success {
        echo 'SUCCESS: Build backup and Tomcat deployment completed.'
    }

    failure {
        echo 'FAILED: Check Console Output before retrying.'
    }
}
}
```



Step 2. Create the backup directories

Run these commands on Ubuntu:

sudo mkdir -p /opt/deploy/tomcat-html/build_backup
sudo mkdir -p /opt/deploy/tomcat-html/deployment_backup

sudo chown -R root:root /opt/deploy/tomcat-html
sudo chmod 755 /opt/deploy
sudo chmod 755 /opt/deploy/tomcat-html
sudo chmod 755 /opt/deploy/tomcat-html/build_backup
sudo chmod 755 /opt/deploy/tomcat-html/deployment_backup

Verify: sudo ls -ld /opt/deploy/tomcat-html/



Step 3. Create the Tomcat application directory

sudo mkdir -p /opt/apache-tomcat-10.1.56/webapps/jenkins-demo
sudo chown -R amit:amit /opt/apache-tomcat-10.1.56/webapps/jenkins-demo
sudo chmod 755 /opt/apache-tomcat-10.1.56/webapps/jenkins-demo

Step 4. Create the deployment script
sudo nano /usr/local/sbin/deploy-jenkins-demo

```
#!/bin/bash
set -Eeuo pipefail

# Expected Jenkins job workspace
EXPECTED_WORKSPACE="/var/lib/jenkins/workspace/jenkins-tomcat-demo"

# Source from Jenkins
SOURCE_DIR="${1:?Workspace argument is required}"
SOURCE_FILE="$SOURCE_DIR/index.html"

# Live Tomcat application
TARGET_DIR="/opt/apache-tomcat-10.1.56/webapps/jenkins-demo"
TARGET_FILE="$TARGET_DIR/index.html"

# Backup directories
BASE_BACKUP="/opt/deploy/tomcat-html"
BUILD_BACKUP="$BASE_BACKUP/build_backup"
DEPLOY_BACKUP="$BASE_BACKUP/deployment_backup"

# Verify that the source is the expected Jenkins workspace
SOURCE_REAL="(realpath"SOURCE_DIR")"
EXPECTED_REAL="(realpath − m"EXPECTED_WORKSPACE")"

if [ "SOURCE_REAL"! = "EXPECTED_REAL" ]; then
    echo "ERROR: Unexpected workspace path."
    exit 1
fi

# Validate source file
if [ ! -s "$SOURCE_FILE" ]; then
    echo "ERROR: index.html is missing or empty."
    exit 1
fi

if ! grep -qi '<html' "$SOURCE_FILE"; then
    echo "ERROR: index.html does not look like an HTML document."
    exit 1
fi

# Unique date-and-time folder for this run
TIMESTAMP="$(date +%Y%m%d_%H%M%S_%N)"
BUILD_DIR="BUILD_BACKUP/TIMESTAMP"
DEPLOY_DIR="DEPLOY_BACKUP/TIMESTAMP"
```

```
echo "Deployment timestamp: $TIMESTAMP"

# 1. Save new HTML in build_backup
mkdir -p "$BUILD_DIR"
install -o root -g root -m 644 \
   "SOURCE_FILE""BUILD_DIR/index.html"

echo "Build backup: $BUILD_DIR/index.html"

# 2. Ensure the live application directory exists
mkdir -p "$TARGET_DIR"

# 3. Back up the old live HTML, if present
if [ -f "$TARGET_FILE" ]; then
   mkdir -p "$DEPLOY_DIR"
   install -o root -g root -m 644 \
      "TARGET_FILE""DEPLOY_DIR/index.html"

   echo "Deployment backup: $DEPLOY_DIR/index.html"
else
   echo "First deployment: no previous index.html."
fi

# 4. Install new HTML using a temporary file and atomic rename
TEMP_FILE="(mktemp"TARGET_DIR/.index.html.XXXXXX")"
trap 'rm -f "${TEMP_FILE:-}"' EXIT

install -o amit -g amit -m 644 \
   "BUILD_DIR/index.html""TEMP_FILE"

mv -fT "TEMP_FILE""TARGET_FILE"
TEMP_FILE=""

echo "SUCCESS: HTML deployed to Tomcat."
echo "Live file: $TARGET_FILE"
```

```
 GNU nano 6.2                                  /usr/local/sbin/deploy-jenkins-demo *
#!/bin/bash
set -Eeuo pipefail

# Expected Jenkins job workspace
EXPECTED_WORKSPACE="/var/lib/jenkins/workspace/jenkins-tomcat-demo"

# Source from Jenkins
SOURCE_DIR="${1:?Workspace argument is required}"
SOURCE_FILE="$SOURCE_DIR/index.html"

# Live Tomcat application
TARGET_DIR="/opt/apache-tomcat-10.1.56/webapps/jenkins-demo"
TARGET_FILE="$TARGET_DIR/index.html"

# Backup directories
BASE_BACKUP="/opt/deploy/tomcat-html"
BUILD_BACKUP="$BASE_BACKUP/build_backup"
DEPLOY_BACKUP="$BASE_BACKUP/deployment_backup"

# Verify that the source is the expected Jenkins workspace
SOURCE_REAL="$(realpath "$SOURCE_DIR")"
EXPECTED_REAL="$(realpath -m "$EXPECTED_WORKSPACE")"

if [ "$SOURCE_REAL" != "$EXPECTED_REAL" ]; then
    echo "ERROR: Unexpected workspace path."
    exit 1
fi

# Validate source file
if [ ! -s "$SOURCE_FILE" ]; then
    echo "ERROR: index.html is missing or empty."
    exit 1
fi

if ! grep -qi '<html' "$SOURCE_FILE"; then
    echo "ERROR: index.html does not look like an HTML document."
    exit 1
fi

# Unique date-and-time folder for this run
TIMESTAMP="$(date +%Y%m%d_%H%M%S_%N)"
BUILD_DIR="$BUILD_BACKUP/$TIMESTAMP"
DEPLOY_DIR="$DEPLOY_BACKUP/$TIMESTAMP"

echo "Deployment timestamp: $TIMESTAMP"

# 1. Save new HTML in build_backup
mkdir -p "$BUILD_DIR"
install -o root -g root -m 644 \
    "$SOURCE_FILE" "$BUILD_DIR/index.html"

^G Help          ^O Write Out     ^W Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo         M-A Set Mark     M-] To Bracket   M-Q Previous     ^B Back          ^◀ Prev Word
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo         M-6 Copy         ^Q Where Was     M-W Next         ^F Forward       ^▶ Next Word
```

Ctrl + X → Y → Enter

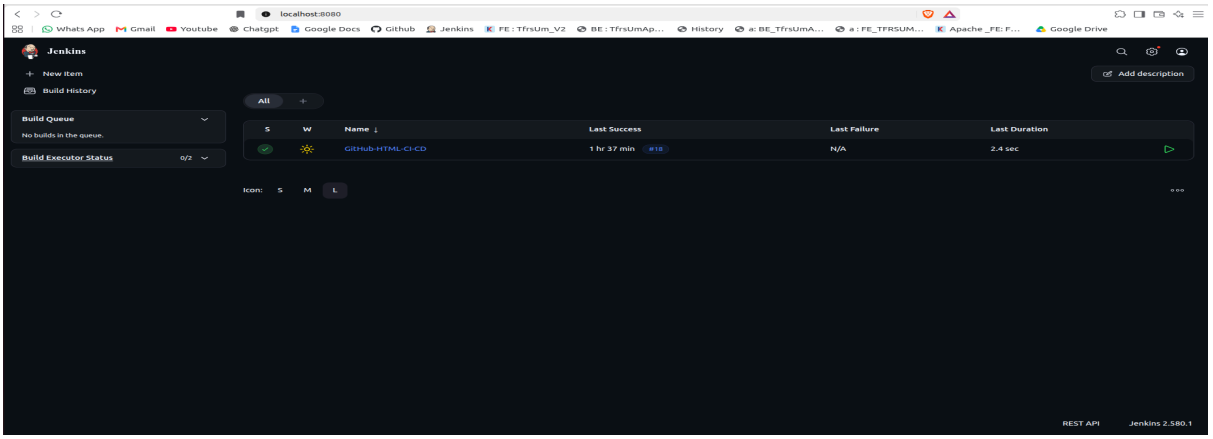
Set its ownership and permissions:
sudo chown root:root /usr/local/sbin/deploy-jenkins-demo
sudo chmod 750 /usr/local/sbin/deploy-jenkins-demo

Step 6. Allow Jenkins to execute only this script
sudo visudo -f /etc/sudoers.d/jenkins-demo

Validate the rule:
sudo visudo -cf /etc/sudoers.d/jenkins-demo

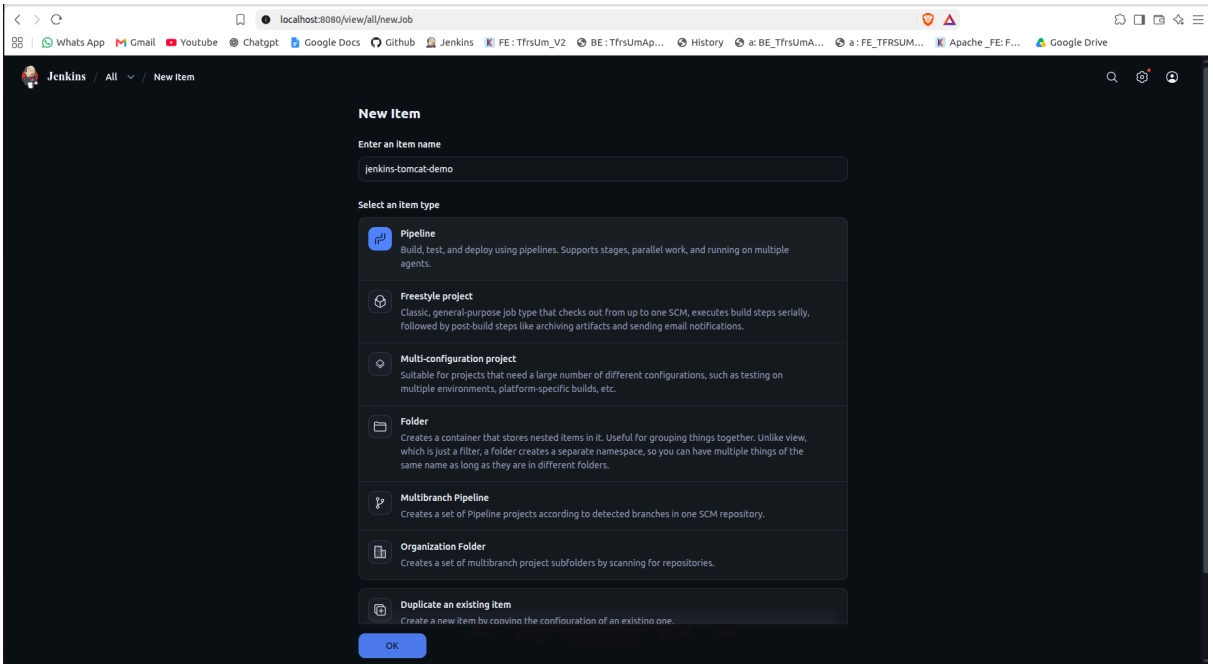
```
amit@red-wolf:/$ sudo visudo -cf /etc/sudoers.d/jenkins-demo
/etc/sudoers.d/jenkins-demo: parsed OK
amit@red-wolf:/$
```

Step 7. Create the Jenkins Pipeline job



1. Create a Jenkins job

Click New Item, enter jenkins-tomcat-demo, select Pipeline, and click OK.



2. Scroll down to the Pipeline section.

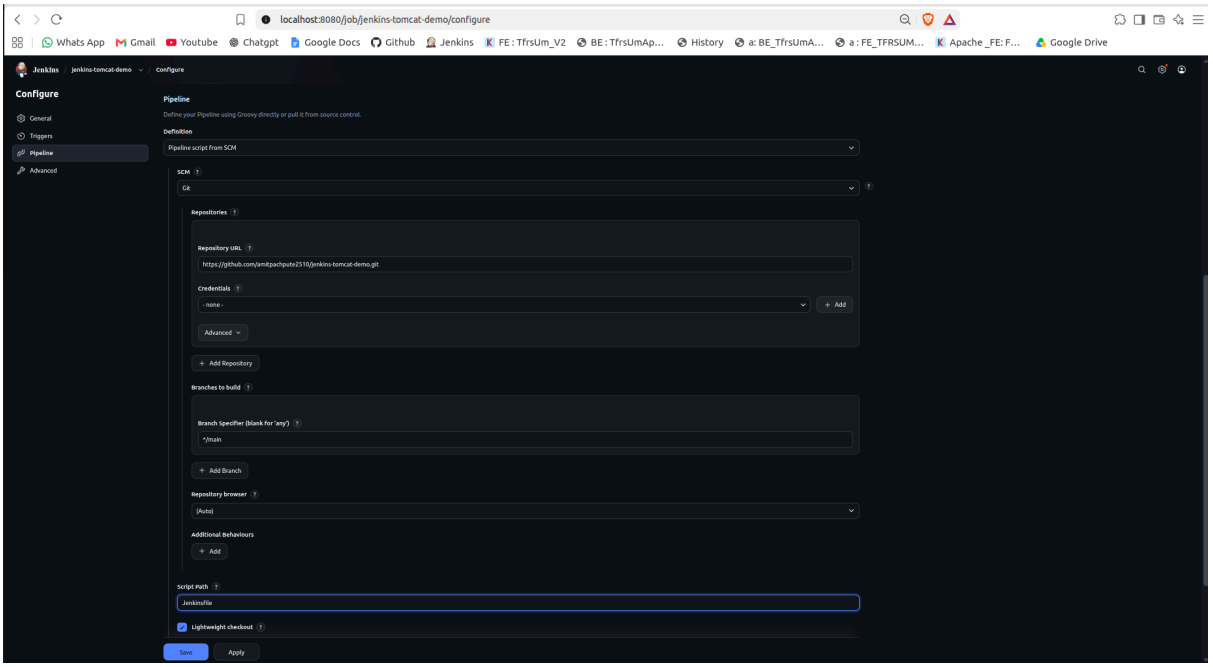
1. Set Definition to Pipeline script from SCM.
2. Select Git as the SCM.

Select Git and enter your repository URL:

https://github.com/YOUR_USERNAME/jenkins-html-demo.git

Replace YOUR_USERNAME with your GitHub username. Set the branch to */main. Leave credentials empty for a public repository. Set Script Path to `Jenkinsfile`

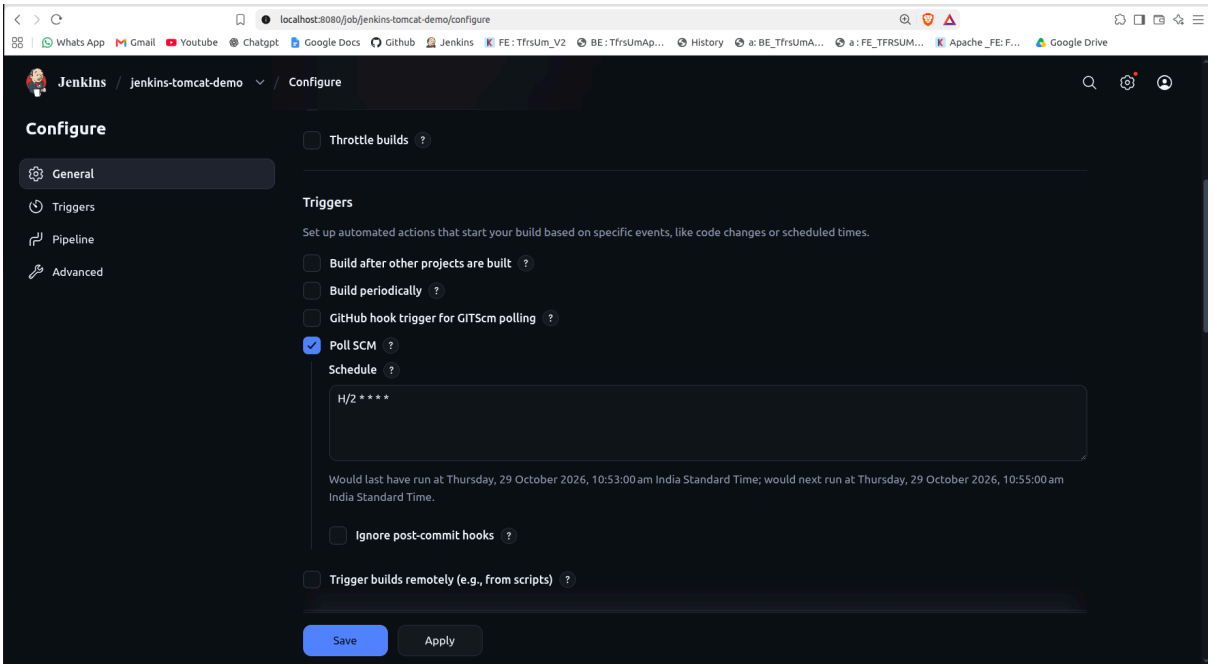
URL : https://github.com/amitpachpute2510/jenkins-tomcat-demo.git



3. Configure automatic polling

Under Build Triggers, select Poll SCM and enter this schedule:

H/2 * * * *



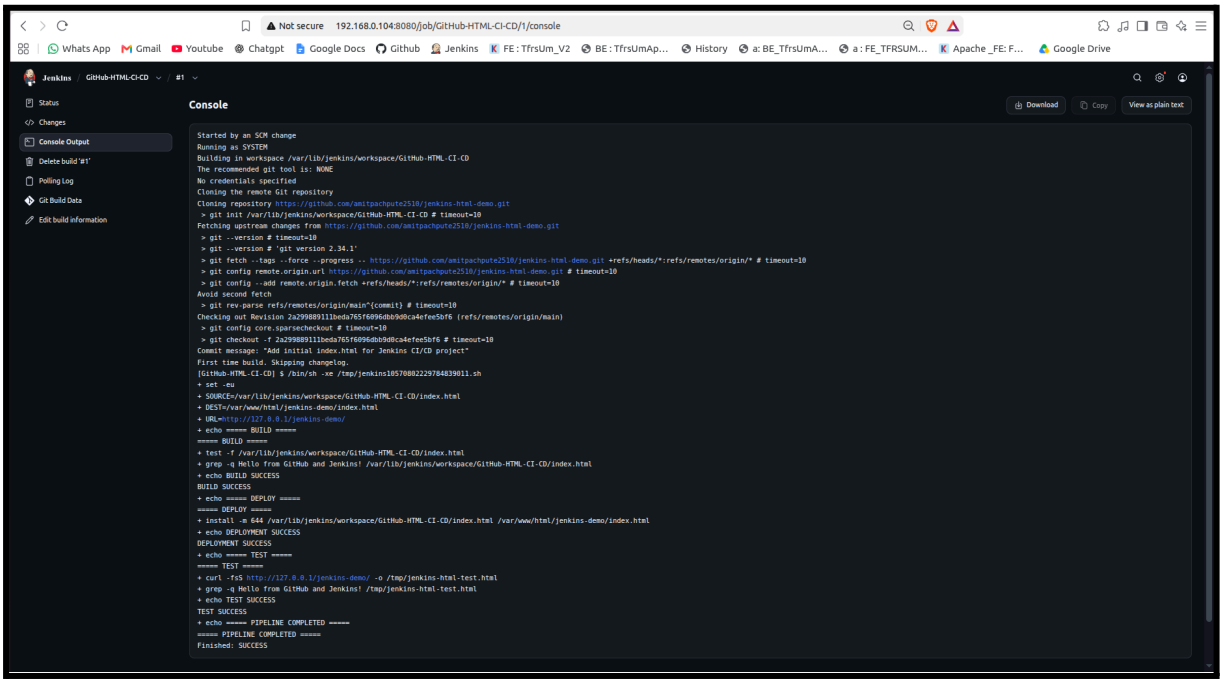
Step 4: There Is Changes

sudo chmod 750 /usr/local/sbin/deploy-jenkins-demo

```
sudo sed -i 's/set -Eeuo pipefail/set -eu/' /usr/local/sbin/deploy-jenkins-demo
sudo head -n 3 /usr/local/sbin/deploy-jenkins-demo
```

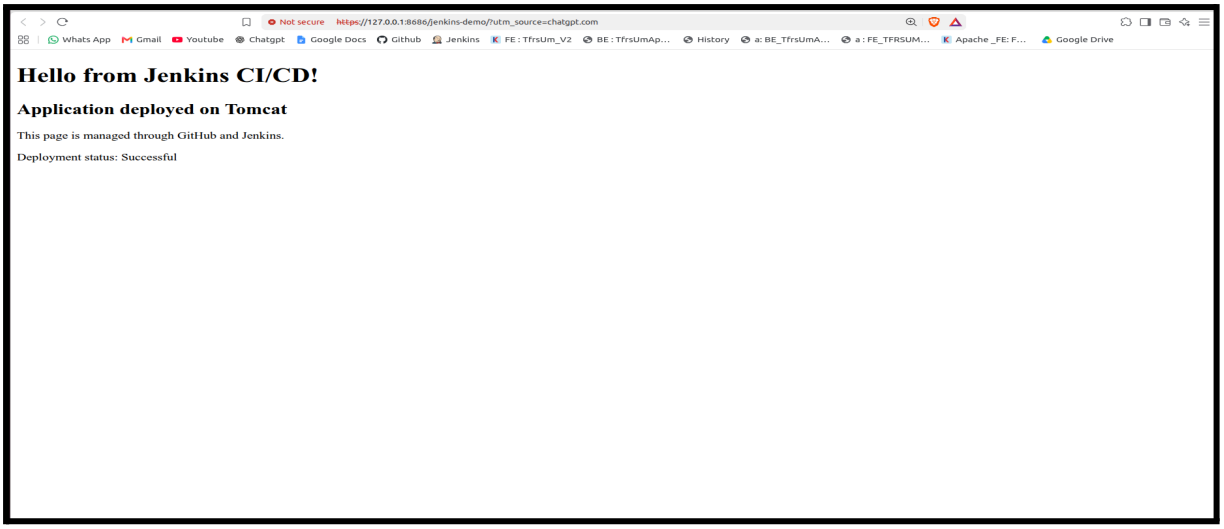
Step 5: Run and verify

1. Click Build Now in your Jenkins job.
2. Open Build History → latest build → Console Output.
3. Confirm the output ends with Finished: SUCCESS.
4. Open your website in a browser.



If your server IP is still 192.168.0.104,

the URL is: http://192.168.0.104/jenkins-demo/



Step 1: Open your project in VS Code

cd /var/lib/jenkins/workspace/jenkins-tomcat-demo
code .

Step 2: Clone your GitHub repository for editing

mkdir -p ~/Projects
cd ~/Projects
git clone https://github.com/amitpachpute2510/jenkins-tomcat-demo.git
cd jenkins-tomcat-demo
code .

cd jenkins-tomcat-demo
code .

